

Simultaneous Recognition of Multiple Signals in Bacterial Outer Membrane Proteins Enhance Assembly and Maintain Membrane Integrity

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Abstract

Outer membrane proteins (OMPs) are essential components of the outer membrane of Gram-negative bacteria. In terms of protein targeting and assembly, the current dogma holds that a “ β -signal” imprinted in the final β -strand of the OMP engages the β -barrel assembly machinery (BAM complex) to initiate membrane insertion and assembly of the OMP into the outer membrane. Here, we reveal an additional rule, that signals equivalent to the β -signal are repeated in other, internal β -strands within bacterial OMPs. The internal signal is needed to promote the efficiency of the assembly reaction of these OMPs. BamD, an essential subunit of the BAM complex, recognizes the internal signal and the β -signal, arranging several β -strands for rapid OMP assembly. The internal signal-BamD ordering system is not essential for bacterial viability but is necessary to retain the integrity of the outer membrane against antibiotics and other environmental insults.

Teaser

Bacterial outer membrane proteins are recognized and bound by BamD at specific signals located in multiple β -strands at the C-terminus of these proteins.

eLife assessment

The biogenesis of outer membrane proteins in Gram-negative bacteria is still not fully understood, particularly substrate recognition and insertion by beta-assembly machinery. This work reports **important** results identifying a new amino acid sequence motif (i.e., "internal beta-signal") on outer membrane proteins recognized by this machinery. The authors carried out rigorous biochemical approaches, providing **convincing** evidence to support their conclusions.

Introduction

In Gram-negative bacteria outer membrane proteins (OMPs) play important roles in membrane-mediated processes (Lundquist et al., 2021 [↗](#); Tommassen, 2010 [↗](#); Walther et al., 2009 [↗](#)). Structurally, each major OMP has a β -barrel transmembrane domain, spanning the membrane as a cylindrical structure formed from amphiphilic, antiparallel β -strands (Konovalova et al., 2017 [↗](#); Schulz, 2000 [↗](#)). Starting as nascent polypeptides synthesized by ribosomes in the cytoplasm, these newly synthesized OMPs cross the inner membrane in a process mediated by the SecYEG translocon (Manting et al., 2000 [↗](#); Mori and Ito, 2001 [↗](#)), and then traverse the periplasm supported by the periplasmic chaperones (Wang et al., 2021 [↗](#)). The fundamental insertase for OMP insertion and assembly into the membrane is the β -barrel Assembly Machinery complex (BAM complex) (Bakelar et al., 2016 [↗](#); Gu et al., 2016 [↗](#); Iadanza et al., 2016 [↗](#)). Given this essential role, and its location on the bacterial cell surface, the BAM complex has emerged as an attractive antibiotic target (Hart et al., 2019 [↗](#); Imai et al., 2019 [↗](#); Kaur et al., 2021 [↗](#); Luther et al., 2019 [↗](#)).

In *E. coli*, the BAM complex is composed of five subunits BamA, BamB, BamC, BamD and BamE (Bakelar et al., 2016 [↗](#); Gu et al., 2016 [↗](#); Iadanza et al., 2016 [↗](#); Wu et al., 2005 [↗](#)). BamA, the essential core-subunit, is embedded in the outer membrane and projects an extensive N-terminal domain consisting of five polypeptide transport-associated (POTRA) repeats into the periplasm which interacts with other subunits (Noinaj et al., 2013 [↗](#); Voulhoux, 2003 [↗](#)). The lipoprotein subunits present in the BAM complex varies in different bacterial lineages (Anwari et al., 2012 [↗](#); Webb et al., 2012 [↗](#)) but both BamA and the lipoprotein BamD are found in the BAM complexes of all bacterial lineages and are essential for viability in *E. coli* (Malinverni et al., 2006 [↗](#); Webb et al., 2012 [↗](#); Wu et al., 2005 [↗](#)). The interaction of the POTRA domains of BamA with its partner BamD contributes to a large, funnel-like feature of the BAM complex which expands into the periplasm.

BamA belongs to Omp85-superfamily protein conserved across bacterial lineages and organelles (Wu et al., 2005 [↗](#)). Omp85-superfamily proteins are themselves β -barrels, utilizing a transiently open lateral gate between the first and last β -strands as the catalytic site of the OMPs assembly (Doyle et al., 2022 [↗](#); Doyle and Bernstein, 2021 [↗](#), 2019 [↗](#); Noinaj et al., 2013 [↗](#); Takeda et al., 2021 [↗](#); Tomasek et al., 2020 [↗](#); Xiao et al., 2021 [↗](#)). Biophysical studies have shown "snap-shot" structures of assembly intermediates of the BAM complex with substrate OMPs and demonstrated that during assembly the first strand of the lateral gate of the BamA interacts with the C-terminal strand of the substrate OMPs (Doyle et al., 2022 [↗](#); Shen et al., 2023 [↗](#); Takeda et al., 2023 [↗](#); Tomasek et al., 2020 [↗](#); Wu et al., 2021 [↗](#)). OMPs contain a conserved sequence termed the " β -signal" located within this C-terminus strand (Kutik et al., 2008 [↗](#); Paramasivam et al., 2012 [↗](#); Struyvé et al., 1991 [↗](#); Wang et al., 2021 [↗](#)). Inhibitory compounds such as darobactin show a similar structure to that of the β -signal and are capable of interacting with the lateral gate (Imai et

al., 2019 [↗](#); Kaur et al., 2021 [↗](#); Xiao et al., 2021 [↗](#)). The current dogma suggested by these studies dictate that this β -signal should engage the first strand of the lateral gate in BamA (Horne and Radford, 2022 [↗](#); Tomasek and Kahne, 2021 [↗](#)).

Despite recent advances in resolving the BAM complex structure and its function, the molecular mechanism of early stage OMP assembly remains enigmatic. It has been shown that the β -signal plays a significant role in the targeting and folding of BAM complex substrates (Kutik et al., 2008 [↗](#); Wang et al., 2021 [↗](#)), yet several important OMPs do not have a recognizable β -signal motif in their C-terminal sequence (Hagan et al., 2015 [↗](#)). For instance, BamA itself lacks an apparent C-terminal β -signal (Hagan et al., 2015 [↗](#)). Also, a comprehensive study of the autotransporter class of OMPs, including proteins such as EspP, showed that while 1038 non-redundant proteins did have a C-terminal β -signal motif, over 40% of these proteins had a β -signal motif only in internal strands (Celik et al., 2012 [↗](#)), which was also suggested to be the case for BamA (Hagan et al., 2015 [↗](#)). It remains to be tested if these internal motifs function as β -signals and how they engage the BAM complex.

Before substrate OMPs can enter the later gate of BamA, they must first pass through the POTRA domain of the BAM complex. Although the assembly reactions that occur at the lateral gate are well understood, the molecular mechanisms at the early stage, prior to being position in the lateral gate, are largely unknown. The specific points that have been established are that the substrate makes contact with the BamB and BamD subunits and the lateral gate of the BamA subunit (Hagan et al., 2013 [↗](#); Harrison, 1996 [↗](#); Ieva et al., 2011 [↗](#)).

Here, we establish a screening system based on an *in vitro* reconstituted membrane assay using a translocation-competent *E. coli* Microsomal Membrane (EMM) (Gunasinghe et al., 2018 [↗](#); Thewasano et al., 2023 [↗](#)) and demonstrate that the majority of OMPs, including the major porins that are the predominant substrate of the BAM complex, contain multiple β -signal related repeats. Comparison of the C-terminal β -signal with what we refer to here as the “internal signal”, revealed the essential elements of sequence and orientation in the signals that are recognized by the BAM complex. Internal signals and β -signals are recognized by BamD and are responsible for rapid assembly of the OMP into the bacterial membrane at the early stage. Moreover, in Gram-negative bacteria, *in situ* cross-linking and mutagenesis demonstrate how BamD simultaneously recognized these two β -signals, within the context of a functional BAM complex.

Results

Peptidomimetics derived from *E. coli* OmpC inhibit OMP assembly

The EMM assembly assay provides a means for *in vitro* reconstitution of BAM complex function, where the assembly of a ^{35}S -labelled OMP can be monitored (fig. S1A) (Gunasinghe et al., 2018 [↗](#); Thewasano et al., 2023 [↗](#)). When applied to an autotransporter like ^{35}S -labelled EspP, the assay discriminates assembly of a protease-protected β -barrel domain, assembled into the OM by the BAM complex (Ieva et al., 2011 [↗](#)), from the protease-sensitive extracellular passenger domain (Fig. 1A [↗](#)) that are degraded by protease (Leyton et al., 2014 [↗](#); Roman-Hernandez et al., 2014 [↗](#)). To screen for potential peptidomimetics that could inhibit BAM complex function, a synthetic peptide library was designed for coverage of the sequence features in OmpC (fig. S2). Under the conditions of the EMM assay, ^{35}S -labelled EspP was observed to fragment as predicted (fig. S1B) and was used to screen inhibitory peptides in a robust high-throughput format.

Six peptides (4, 10, 17, 18, 21, and 23) were found to inhibit EspP assembly (Fig. 1A [↗](#)). Five of the most potent peptides (4, 17, 18, 21, and 23) also inhibited additional model OMPs; the porins OmpC and OmpF, the peptidoglycan-binding OmpA, and the maltoporin LamB (fig. S3). Comparing the sequences of these inhibitory peptides suggested the presence of a $[\Omega/\Phi]_x[\Omega/\Phi]$ motif (Fig. 1B [↗](#)),

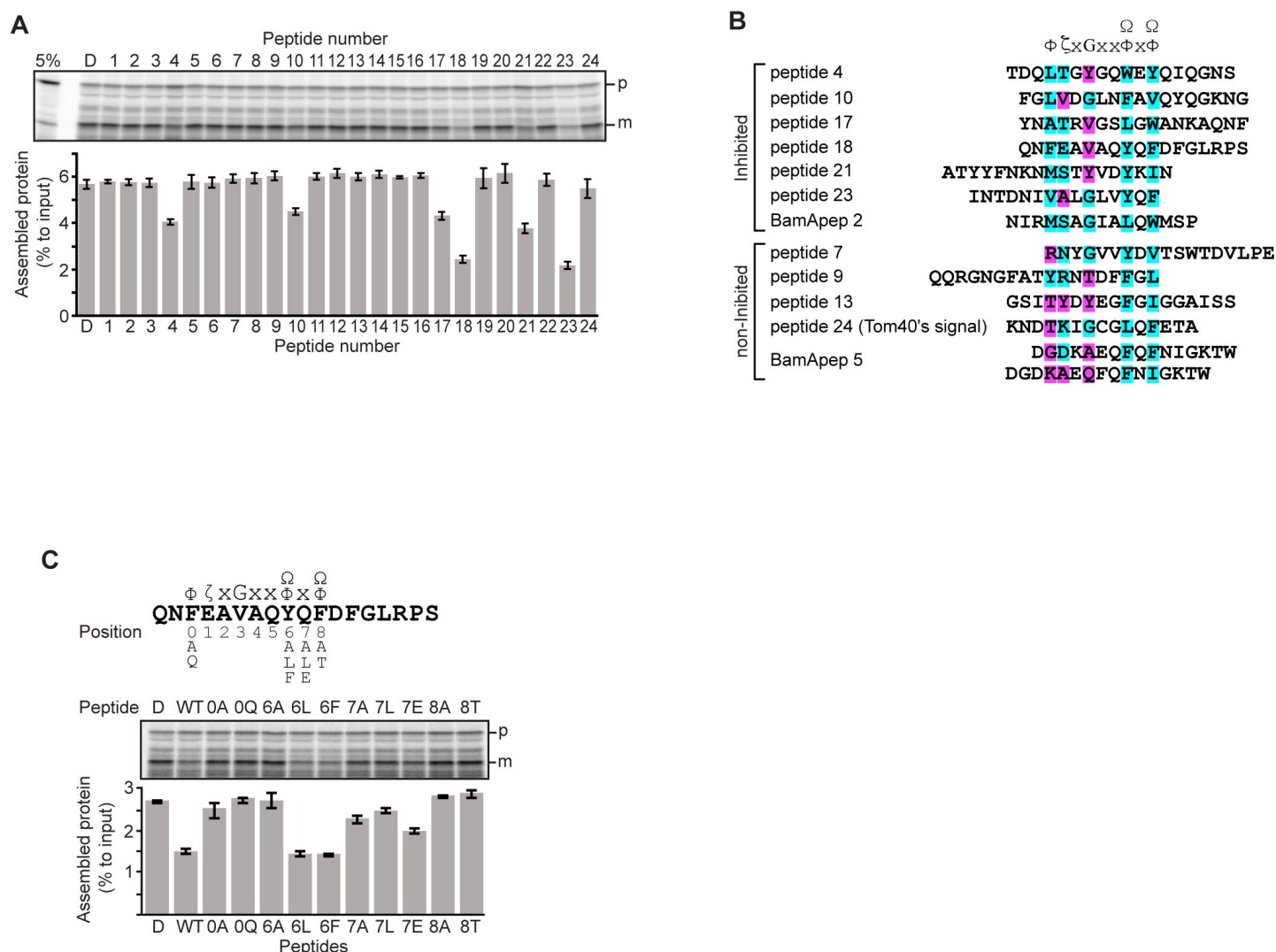


Fig. 1 |

Peptides derived from β -strands of OmpC selectively inhibit OMP assembly.

A and C, ^{35}S -labelled EspP was synthesized by rabbit reticulocyte lysate, then incubated with EMM, and samples prepared for analysis by SDS-PAGE and radio-imaging (see Methods). EspP assembly into EMM was performed in the presence of DMSO (D) or specific inhibitor peptides. The precursor (p) and mature (m) form of EspP are indicated. **A**, Peptides labelled 1 through 23 were derived from OmpC, while peptide 24 was taken from a mitochondrial β -barrel protein to serve as a negative control (see fig. S2). **B**, Sequence comparison of inhibitor peptides identified in the EspP assembly assays. The signal is comprised of key hydrophobic (Φ), polar (ζ), glycine (G), and 2 aromatic (Ω) or hydrophobic amino acids. Amino acids highlighted in blue conform to the consensus signal, while those highlighted in purple deviate from the consensus signal. **C**, Upper: sequence of the peptide 18 indicating residue codes for the mutated peptides e.g. at position "6", the native residue Y was mutated to A in peptide "6A". Position 0 designates the first hydrophobic residue and position 8 designates the final aromatic or hydrophobic residue. DMSO (D) was used for control. Middle: SDS-PAGE of EspP assembly assay. Lower: data for inhibition of EspP assembly into EMM assayed in the presence of each of the indicated peptides.

which is part of the motif documented in the β -signal (ζ xGxx[Ω/Φ]x[Ω/Φ]). The sequence codes refer to conserved features in residues such that: ζ , is any polar residue; G is a glycine residue; Ω is any aromatic residue; Φ is any hydrophobic residue and x is any residue (Hagan et al., 2015; Kutik et al., 2008). The non-inhibitory peptide 9 contained some elements of the β -signal but did not show inhibition of EspP assembly (Fig. 1A).

Peptide 18 showed a strong sequence similarity to the motif (Fig. 1B) and also had a strong inhibitory action on EspP assembly (Fig. 1A), so variant peptides based on the peptide 18 sequence were constructed and tested in the EMM assembly assay (Fig. 1C). This analysis revealed that the position 0 (0Φ), in addition to the [Ω/Φ]x[Ω/Φ], motif was functionally important to inhibitory action (Fig. 1C). Thus, we hypothesized the elements of an internal β -signal (hereafter the internal signal) to be Φ xxxxx[Ω/Φ]x[Ω/Φ].

Mutations to a putative internal signal results in loss of OMP assembly

To assess the impact of the conserved residues in the potential internal signal for OMP assembly, a series of mutant OmpC variants were constructed and assayed for assembly in the EMM assay (Fig. 2A). Three of the peptide regions are present in the final 5 β -strands of OmpC. We denoted these as the first (-1), third (-3), and fifth (-5) β strands from the C-terminus of the barrel. Note, since peptide 17 and peptide 18 overlap, they were considered together as a single region. Given the antiparallel structure of a β -barrel the -1, -3 and -5 strands are orientated in the same direction, and these three strands all conform to the putative Φ xxxxx[Ω/Φ]x[Ω/Φ] motif.

BN-PAGE analysis was employed to measure the efficiency of OmpC trimer assembly (Fig. 2A). These gels separate membrane proteins that have been solubilized from native membranes, allowing time-course analysis of protein assembly. Among the mutants with significantly reduced assembly efficiency, Y286A, Y325A, and Y365A correspond to Ω/Φ in position 6 (Ω/Φ), and F280A corresponds to Φ in position 0 of the putative signal (Fig. 2A and fig. S4A). In terms of the highly conserved residues present, sequence conservation was mapped against the β -strands of OMPs whose structures have been determined (fig. S5, S6, and S7). Analysis showed the -5 strand of OmpF and LamB includes residues that conform to the putative Φ xxxxx[Ω/Φ]x[Ω/Φ] motif (Fig. 2B). We constructed mutants to the 0Φ or 6[Ω/Φ] positions of the -5 strand in OmpF and LamB, these are the OmpF(V279A), OmpF(Y285), LamB(V333A) and LamB(Y339A) mutants (Fig. 2B). BN-PAGE analysis showed that both the Φ - OmpF(V279) and LamB(V333) - and Ω/Φ - OmpF(Y285) and LamB(Y339) - residues were important for the assembly of OmpF and LamB (fig. S4B, C).

To address whether the results seen *in vitro* in the EMM assays were consistent with phenotypes drawn from intact *E. coli* cells, we constructed arabinose-inducible plasmids to express mutants of Φ (position 0), the [Ω/Φ] (position 6) and a double mutant combination of both positions in either the -5 or -1 strands of OmpC. To distinguish these plasmid-encoded OmpC variants from the endogenous, chromosome-encoded OmpC, a FLAG epitope tag was incorporated into the region of OmpC immediately behind the cleavage site for the secretory signal sequence (Fig. 3A) (Rapoport, 2007). Expression of FLAG-OmpC variants was induced with 0.1% arabinose, as confirmed through immunoblotting of total cell lysate (Fig. 3B). In comparing the *E. coli* strains, a marked reduction was seen in the steady-state level of the double ("FY" or "VY") mutants OmpC (F280A+Y286A or V359A+Y365A) (Fig. 3C). Little change was evident in the expression levels of the single -1 mutants, suggesting that a fully conserved β -signal located at the C-terminus alone is not essential to maintaining OmpC assembly. To further investigate the nature of the defects, the various strains were analyzed BN-PAGE. In all cases, OmpC that has been detergent-solubilized from outer membranes migrates as a characteristic population of dimeric and trimeric forms under conditions of the BN-PAGE (Fig. 3D). Both the trimeric and dimeric forms of OmpC solubilized from the β -barrel arrays in the outer membrane have equivalent structural and functional properties (Rocque and McGroarty, 1989). The double ("FY" or "VY") mutants

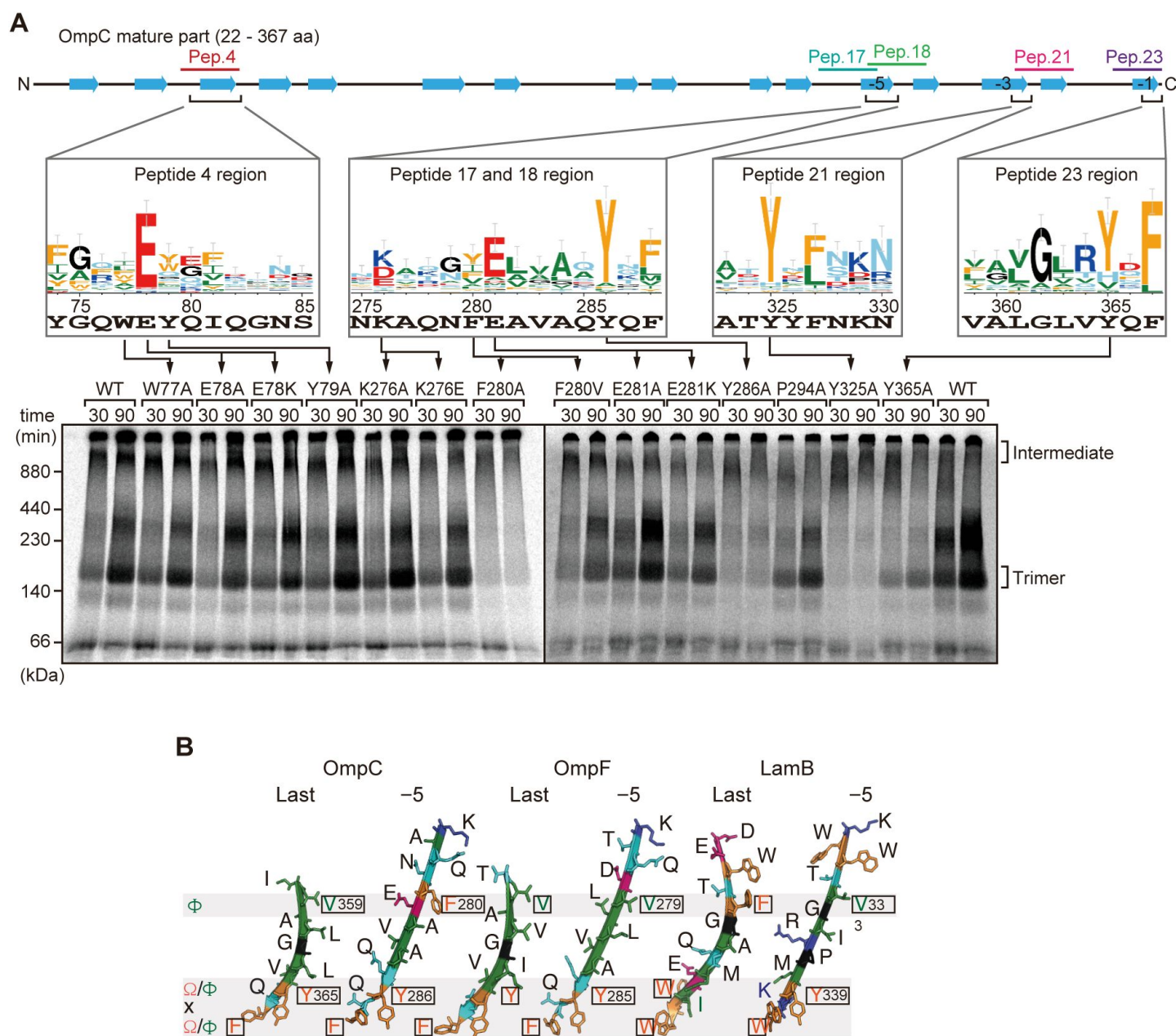


Fig. 2 |

Mutations in β -signals inhibit OMP Assembly *in vitro*.

A, Representation of OmpC, (upper panel) annotated to show the position of each β -strand (blue arrows) and the region mimicked in the indicated peptides. Sequence logos show the conservation of sequence for the residues in these regions (see Methods), where the height of letter corresponds to the degree of conservation across bacterial species, residues are color-coded: aromatic (orange), hydrophobic (green), basic (blue), acidic (red), polar (sky blue) and proline and glycine (black). In the lower panel, data from the EMM assembly assay of OmpC mutants is organized to show effects for key mutants: ^{35}S -labelled OmpC wild-type or mutant proteins were incubated with EMM for 30 or 90 min, and analyzed by BN-PAGE and radio-imaging. **B**, Comparison of amino acids in the final and 5th to final (-5) β -strands of OmpC, OmpF, and LamB. All three OMPs contain the putative hydrophobic residue near the N-terminal region of the β -strands, highlight with a black box. A hydrophobic and an aromatic residue is also found to be highly conserved between these OMP β -strands at the C-terminal amino acids of the β -strands.

displayed drastic reductions of the dimer and trimer (**Fig. 3D**). As a control, the level and integrity of the BAM complex were monitored and found to be equivalent in all strains (**Fig. 3C**, and **D**). Taken together these results suggest either the internal signal (in strand -5) or the C-terminal β -signal (in strand -1) must be present to achieve efficient assembly of OmpC into the outer membrane.

The internal signal is necessary for insertion step of assembly into OM

The β -signal located in the C-terminal β -strand is important for recognition by the BAM complex. In addition, because it is the last β -strand, it also contributes to engagement with the first β -strand in order that purified OMPs can fold into β -barrels, even in the absence of the BAM complex in artificial non-membrane environments such as detergent micelles (Burgess et al., 2008). To compare this property of strand-engagement for β -barrel folding in detergent micelles, we purified urea-denatured OmpC and the OmpC variant proteins with mutations in the -5 strand (F280A and Y286A) and the -1 strand (V359A and Y365A). The spontaneous folding of each OmpC variant was assessed by the rate of formation of trimer in the presence of detergent micelles (fig.S8A). Trimer embedded into detergent micelles migrated at 100 kDa in this assay system, whereas non-folded monomeric OmpC migrated at 37 kDa (fig. S8). Relative to the rate/extent of trimer formation observed for OmpC(WT), mutations to the $[\Omega/\Phi]$ (position 6) in the -5 strand i.e. OmpC(Y286A) or the -1 strand i.e. OmpC(Y365A) showed significantly impaired spontaneous folding. Mutation of the Φ residue (position 0) in the -5 strand i.e. OmpC(F280A) or the -1 strand i.e. OmpC(V359A) maintained a similar ability to assemble into micelles as native OmpC. Thus, the conserved features of both -1 and -5 strands have equivalent impact on intrinsic properties for spontaneous folding of β -barrel of OmpC.

Compared to the rate at which the trimeric form of OmpC is assembled (**Fig. 4A**), OmpC(Y286A) is slow and the gels resolve that most of the ^{35}S -labelled OmpC(Y286A) is held in a high-molecular-weight species for most of the time-course. Only at 80 min of assembly has substantial amount of the ^{35}S -labelled OmpC(Y286A) been resolved from this assembly intermediate (“Int”) to a native, trimeric form (**Fig. 4A**). In analyzing assembly intermediates of other sorts of membrane proteins in mitochondria, a version of this analysis referred to as “gel shift” BN-PAGE analysis has been developed (Shiota et al., 2012). Here, the addition of an antibody recognizing the surface-exposed BamC to the samples prior to electrophoresis, resulted in a dramatic shift (i.e. retardation of electrophoretic mobility due to the added mass of antibodies) of the intermediate, consistent with OmpC(Y286A) being held in an assembly intermediate engaged with the BAM complex (**Fig. 4B**). Consistent with this, the OmpC(Y286A) assembly intermediate can be extracted from the membranes with urea; urea extraction of the membrane samples prior to analysis by BN-PAGE showed that the integral membrane trimer form of OmpC and OmpC(Y286A) were not extracted by urea (**Fig. 4C**). Conversely, the assembly intermediate was completely extracted with urea, as expected for a substrate engaged in protein-protein interactions with a proteinaceous complex in the membrane fraction. These results suggest that the mutation in OmpC(Y286A) causes a longer interaction with the BAM complex, as a species of protein that is not yet inserted into the surrounding membrane environment.

The two essential subunits of the BAM complex are BamA and BamD, and *in vitro* binding assays showed that both BamA and BamD can independently bind to the OmpC polypeptide (fig. S9). To test for signal-dependent binding, BamA and BamD were pre-incubated with signal containing inhibitor peptides (**Fig. 5A**). Peptide 23 inhibited binding of the substrate by BamA and peptide 18 inhibited binding of the substrate by BamD. Peptide 21 inhibited binding of the substrate by either BamA or BamD. Neutron reflectometry (NR) is a powerful tool for probing the details of substrate-chaperone interactions, and is ideally suited for membrane-embedded proteins (Shen et al., 2014). To apply NR imaging to monitor the engagement of OmpC with the BAM complex,

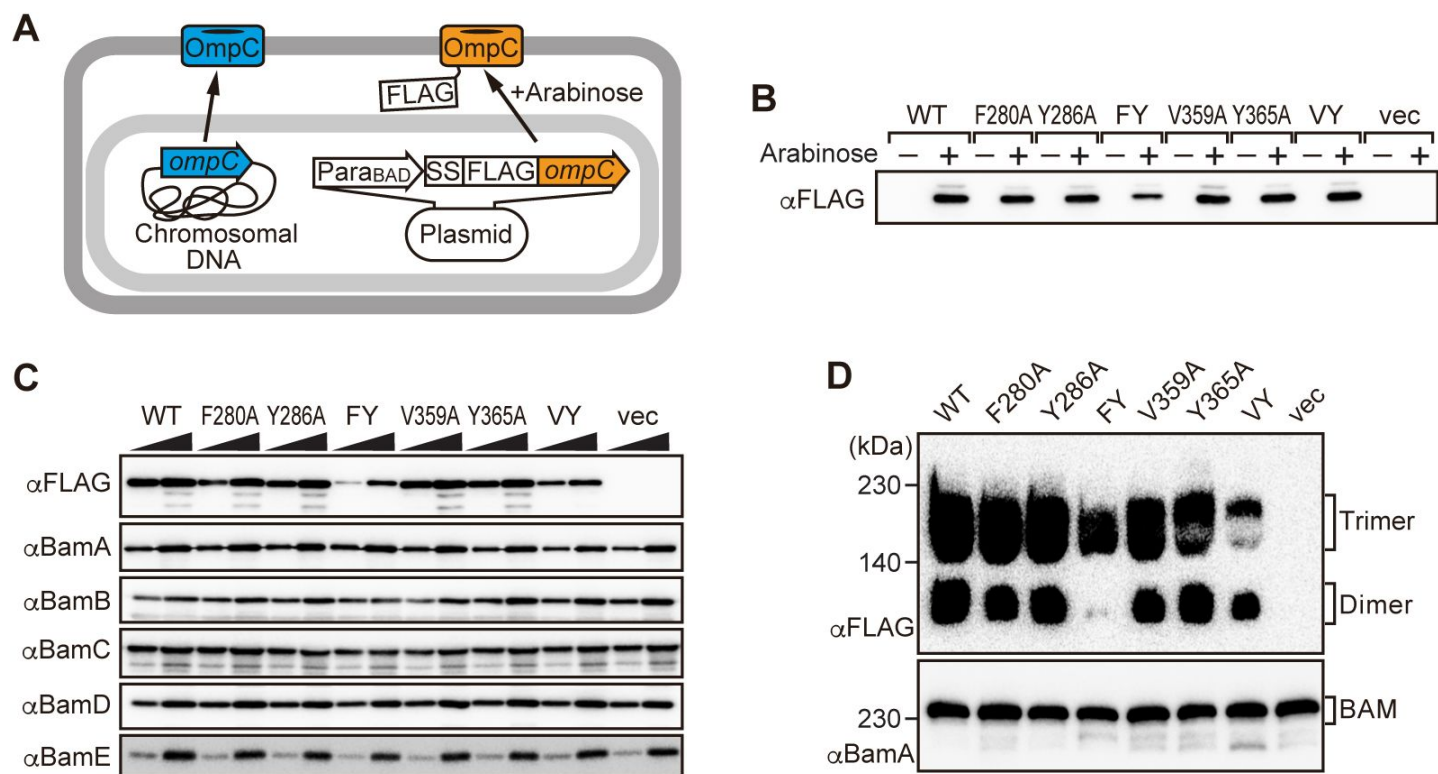


Fig. 3 |

Loss of the putative internal signal disrupts OMP assembly *in vivo*.

A, Schematic model of FLAG-OmpC expression system in *E. coli*. The plasmid-borne copy of OmpC was mutated and the mutant proteins are selectively detected with antibodies to the FLAG epitope. **B**, Total cell lysates were prepared from the variant *E. coli* strains expressing FLAG-OmpC (WT), -5 strand mutants: F280A, Y286A, double mutant F280A/Y286A (FY), or final β -strand mutants: V359A, Y365A, double mutant V359A/Y365A (VY) with (+) or without (-) arabinose. The steady-state levels of the indicated proteins were assessed after SDS-PAGE by immunoblotting with anti-FLAG antibody. **C**, After growth with arabinose induction, EMM fraction was isolated from each of the FLAG-OmpC variant expressing *E. coli* strains to assess steady state levels of OmpC variants and BAM complex subunits: BamA, BamB, BamC, BamD, and BamE by SDS-PAGE and immunoblotting. **D**, EMM fractions were solubilized with 1.5% DDM and subjected to BN-PAGE and immunoblotting. The indicated proteins were detected by anti-FLAG (top) and anti-BamA antibodies (bottom), respectively.

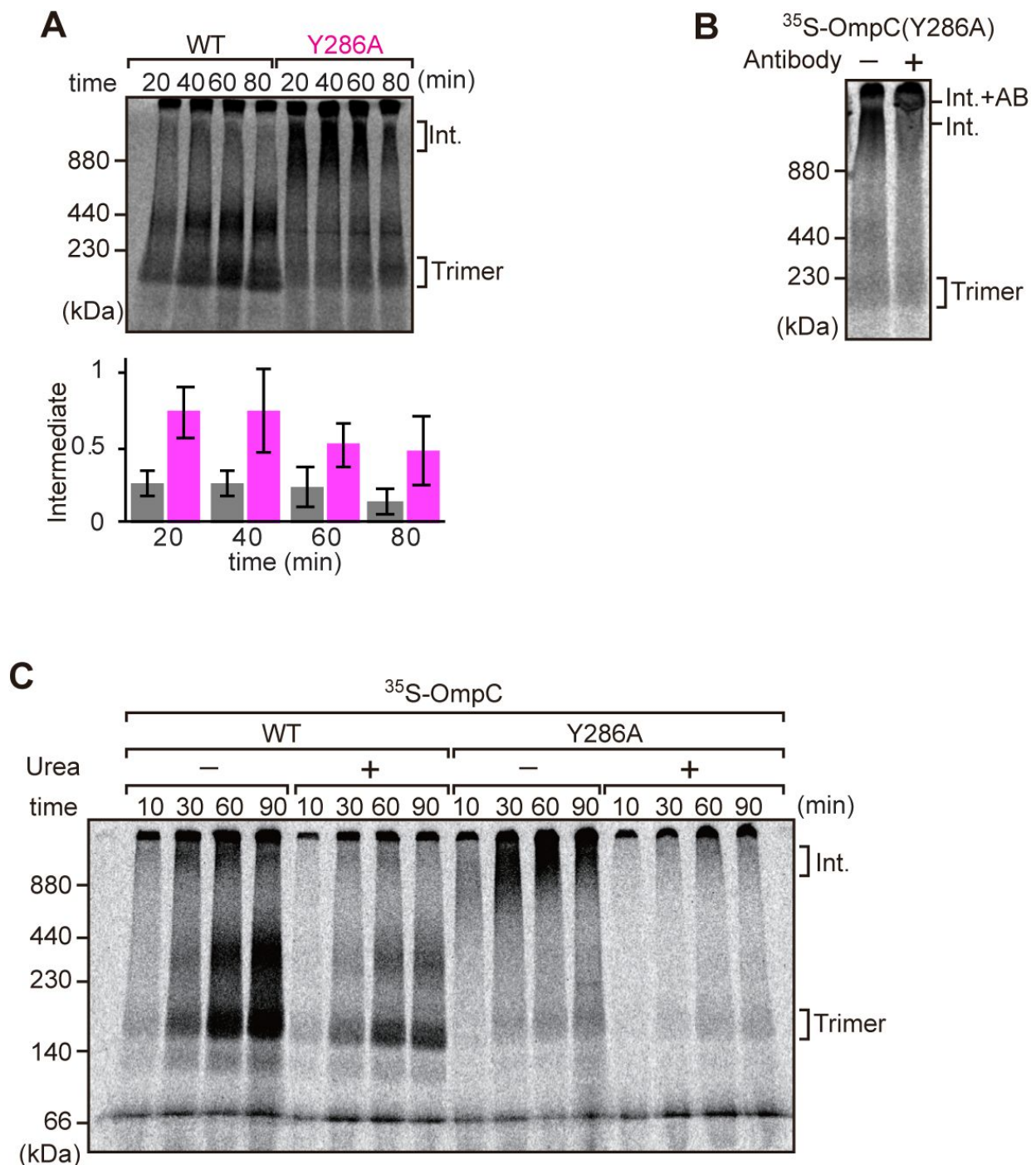


Fig. 4 |

Mutation that stalls OMP assembly in a high molecular weight assembly intermediate.

A, Upper panel, assembly assay of ³⁵S-OmpC wild-type (WT) and mutant Y286A. After incubation in the EMM assays for indicated time points, ³⁵S-proteins were resolved via BN-PAGE. OmpC trimer and intermediate (int) are indicated. The lower panel shows a quantification of the intermediate form compared to total assembled protein (average of 3 independent experiments). **B**, ³⁵S-labelled OmpC(Y286A) was incubated with EMM for 60 mins, then the EMM fraction was solubilized with 1.5% DDM and incubated with (+) or without (-) anti-BamC antibody for 40 min. These samples were then analyzed by BN-PAGE. The presumptive translocation intermediate (Int) and the gel-shift species (Int. +AB) are indicated. **C**, ³⁵S-labelled OmpC was incubated with EMM and then analyzed by BN-PAGE and radio-imaging. EMM were then washed and solubilized in the presence (+) or absence (-) of 6 M urea, after which the membrane fraction was re-isolated via ultracentrifugation for 45 min. Isolated membranes were solubilized in 1.5% DDM and analyzed by BN-PAGE.

BamA was immobilized on an Ni-NTA atomic flat gold-coated silicon wafer to orient the POTRA domain distal to the silicon wafer as previously described (Ding et al., 2020). A lipid layer was reconstituted to provide BamA with a membrane-like environment prior to sequential measurement of: (i) BamA with lipid (1st measurement), (ii) BamA with lipid and BamD (2nd measurement), and (iii) BamA-BamD-lipid with the addition of OmpC (WT) or OmpC(Y286A) (3rd measurement) (fig. S10A).

Comparing the reflectivity profiles of three measurements (Fig. 5B), changes were observed in the layers after the addition of BamD and OmpC(Y286A), as seen in a shift of fringe to low Q range around 0.02-0.03 Å⁻¹. Crystal structures show that the periplasmic domains of BamA can be treated as two rigid bodies: POTRA1-POTRA2(P1-2) and POTRA3-POTRA4-POTRA5(P3-5). Thus, for further data analysis, we analyzed BamA as four layers: the His6 extra-cellular layer where BamA is attached to the chip, the membrane layer (containing the β-barrel domain of BamA), an adjoining P3-5 layer, and the N-terminal P1-2 layer. NR data can then highlight variations in thickness of these layers, which corresponds to the position of additional protein mass in each point of measurement (Fig. 5C, fig. S10B,C).

On addition of BamD (2nd measurement), the SLD profiles showed that BamD was located in the P3-5 layer (Fig. 5C, fig. S10D, E), an observation consistent with previous studies on the resting BAM complex (Ding et al., 2020). On addition of the OmpC(Y286A) substrate (3rd measurement) binding was observed to the P3-5 layer, with a concomitant extension of this layer from 29.7±0.8 Å to 45.5±1.6 Å away from the membrane (Fig. 5C, 5D, fig. S10F,G). By contrast, no obvious changes were observed on the addition of the native OmpC protein implying that interactions it makes with BamA or BamD on the Si-Wafer were resolved too quickly to capture (fig. S11). The failure to observe changes with the WT-OmpC control in the NR analysis is consistent with the data from the BN-PAGE analysis (Fig. 4A) where native OmpC does not form the stable intermediate seen for OmpC(Y286A). The NR findings indicate that OmpC(Y286A) is stacked at a position in the BAM complex with the POTRA3-5 of BamA and BamD, and that the Y286A mutation results in a relatively stable binding to this specific periplasmic region of the BAM complex.

BamD acts as a receptor for the internal signal and β-signal

To elucidate the residues within BamD that mediate this signal-receptor interaction, we applied *in vitro* site-specific photo-crosslinking (fig. S12A). A library of 40 BamD variants was created to incorporate the non-natural amino acid, *p*-benzoyl-L-phenylalanine (BPA), by the suppressor tRNA method in a series of positions (Chin et al., 2002). The UV-dependent appearance of crosslinks between OmpC and BamD occurred for 17 of the BamD variants (fig. S12B,C). The analysis was repeated to determine BamD cross-links for ³⁵S-labelled OmpC(WT), the internal signal mutant OmpC(Y286A), or the β-signal mutant OmpC(βAAA: Y365A, Q366A, F367A) (fig. S12D,E). This mapping of OmpC binding showed that an N-terminal section of BamD (amino acid residues 60 or 65) interacts with the internal signal, while a distinct section of BamD (amino acid residues 196, 200, 204) interacts with β-signal (fig. S12F).

While *in vitro* site-specific photo-crosslinking provides information on the substrate receptor region of BamD, this data only indirectly measures which residues of OmpC are being recognized. In order to obtain complimentary data on OmpC binding by BamD, we purified a form of BamD that is substituted with a cysteine residue in the substrate-binding region, and synthesized forms of ³⁵S-labeled OmpC substituted with a cysteine residue at various positions. Treatment with 1 μM CuSO₄ can stimulate disulfide formation between BamD and OmpC (Fig. 6A). Consistent with the BPA crosslinking data, the positions near the internal signal in OmpC (amino acid residues 278, 284, 296 and 302) formed disulfide bonds with the N-terminal section of BamD (amino acid residues 49, 65). Conversely, residues towards the C-terminal, canonical β-signal of OmpC (amino acid residues 333, 357, and 363) formed disulfide bonds with a C-terminal section of BamD (at amino acid residue 204) (Fig. 6B). All of disulfide crosslinked products were validated by reducing condition SDS-PAGE (fig. S13).

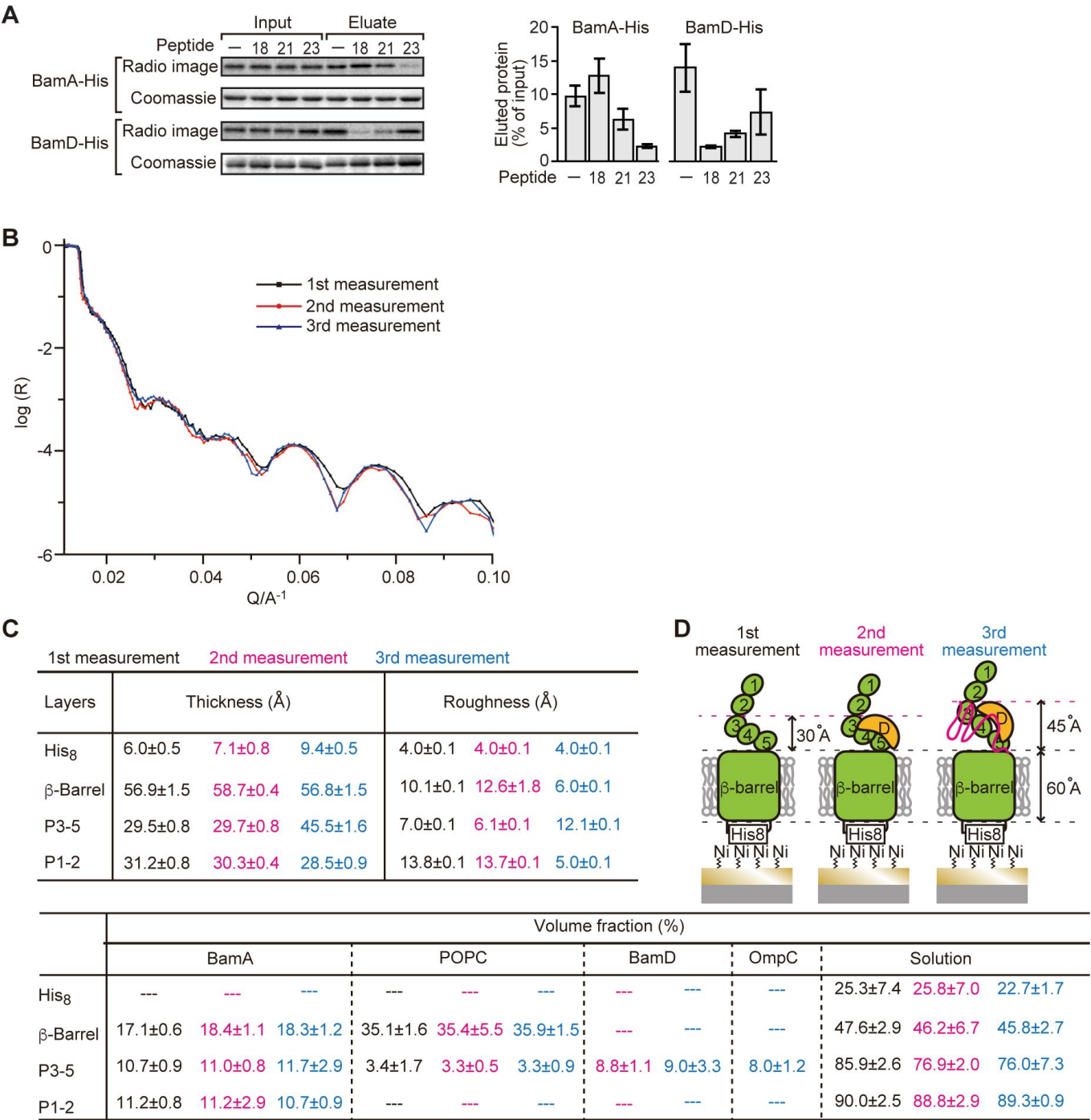


Fig. 5 |

Proximity and dependence of BamD with OmpC(Y286A) revealed by Neutron Reflectometry analysis.

A, Ni-NTA was used to affinity purify His6-BamA or His6-BamD. The BAM complex protein-Ni-NTA bead complex were then incubated in the presence of DMSO or the indicated peptides resuspended in DMSO, and ³⁵S-labelled OmpC was incubated with these beads. After washing, bound proteins were eluted with 400 mM imidazole buffer. The graph represents densitometry analysis of data from 3 independent experiments. **B**, NR profiles for BamA in membrane (1st measurement, square black line), after addition of BamD (2nd measurement, circle red line), and after addition of the OmpC(Y286A) substrate (3rd measurement, triangle blue line). **C**, Table summarizes of the thickness, volume fraction, and roughness of NR analysis. 1st, 2nd, and 3rd measurement displayed in black, magenta, and blue, respectively. P3-5: POTRA3, POTRA 4, and POTRA5, P1-2: POTRA1 and POTRA2, Lipid: POPC, Solution: D₂O, gold match water (GMW) and H₂O. Details were described in Table S8 to S10. **D**, Neutron reflectometry schematic of BamA (green), BamD (yellow) and OmpC(Y286A) (magenta). Numbers indicate corresponding POTRA domains of BamA.

To further probe this interaction, we engineered an *E. coli* strain to express BPA-containing BamD and a FLAG-tagged form of OmpC and performed *in vivo* photo-crosslinking (Fig. 6C). BamD with BPA at positions 49, 53, 65, or 196 were crosslinked to FLAG-OmpC. Positions 114 and 200 in BamD were not detected to interact with OmpC. Structurally, positions 49, 53, 65, or 196 are facing the interior of the funnel-like structure of the periplasmic domain of the BAM complex.

Two structurally distinct regions of BamD promote OmpC folding and assembly

In order to directly measure how the β -strands of OmpC come together, we established a new assay in which the folding of purified OmpC can be measured with or without assistance by purified BamD. Using the crystal structure of OmpC as a guide, cysteine residues were engineered in place of amino acids that were outward-facing on the final five β -strands of folded OmpC, placed close enough to a cysteine residue in the adjacent β -strand to allow for the possibility of disulfide formation once the strands had been correctly aligned in the folded protein (Fig. 7A). These cysteine-variant OmpC proteins were incubated with or without BamD to determine how efficiently disulfide bonds formed. Samples harboring cysteine residues in OmpC that were incubated with BamD migrated faster than non-Cys OmpC on non-reducing SDS-PAGE (Fig. 7B). This migration difference was not observed in reducing conditions, validating that the faster migration occurred as a result of internal molecule disulfide bond formation. Disulfide bonds were formed in each of the paired strands β -strands from -1 to -5, suggesting that BamD directly stimulates arrangement of C-terminal five strands of BamD.

Cross-linking studies shown in Fig. 6B indicated 2 distinct regions of BamD as hot-spots for OmpC-binding. Anticipating that binding sites would be conserved; we analyzed sequence conservations and found BamD(Y62) and BamD(R197) as good candidates proximal to OmpC binding sites (Fig. 7C). Mutant strains expressing BamD(Y62A) and BamD(R197A) in a genetic background where the chromosomal bamD can be depleted by an arabinose-responsive promoter were tested (Fig. 7D and fig. S14A,B,C). Growth conditions were established wherein we could deplete endogenous BamD but not disturb cell growth (fig. S14D, fig. S15A). Isolated EMM from this time point showed that neither the BamD(Y62A) nor the BamD(R197A) mutants affected the steady-state protein levels of other components of the BAM complex (fig. S15B), nor did they impact the complex formation of the BAM complex (Fig. 7E, fig. S15C). The EMM assembly assay showed that the internal signal binding site was as important as the β -signal binding site to the overall assembly rates observed for OmpC (Fig. 7F), OmpF (fig. S15D), and LamB (fig. S15E). These results suggest that recognition of both the C-terminal β -signal and the internal signal by BamD is important for efficient protein assembly.

BamD is responsible for OMP assembly to retain outer membrane integrity

In *E. coli*, BamD function is essential for cell viability (Onufryk et al., 2005). As a result, in an *E. coli* strain where bamD expression is under the control of an arabinose-inducible promoter, in a control strain (Fig. 8A, “vec”) two rounds of dilution in restrictive growth media depletes the level of BamD protein and stops cell growth (Fig. 8A). The BamD(Y62A) mutant and the BamD(R197A) mutant supported viability (Fig. 8A, fig. S15A), allowing membrane fractions to be isolated from each mutant strain: the steady-state level of porins OmpA and OmpC/OmpF was reduced in the BamD(Y62A) and BamD(R197A) mutants (Fig. 8B). This was also true when the level of trimeric OmpF was assessed by native PAGE: less OmpF trimer had been assembled over the growth period relative to the steady-state level of the BAM complex (Fig. 8C). Thus, while these mutant forms of BamD support cell viability (Fig. 8A), they have deleterious effects on β -barrel protein assembly *in vivo*. In the case of the BamD(Y62A) strain, growth in the presence of vancomycin showed impaired outer membrane integrity (Fig. 8D,8E).

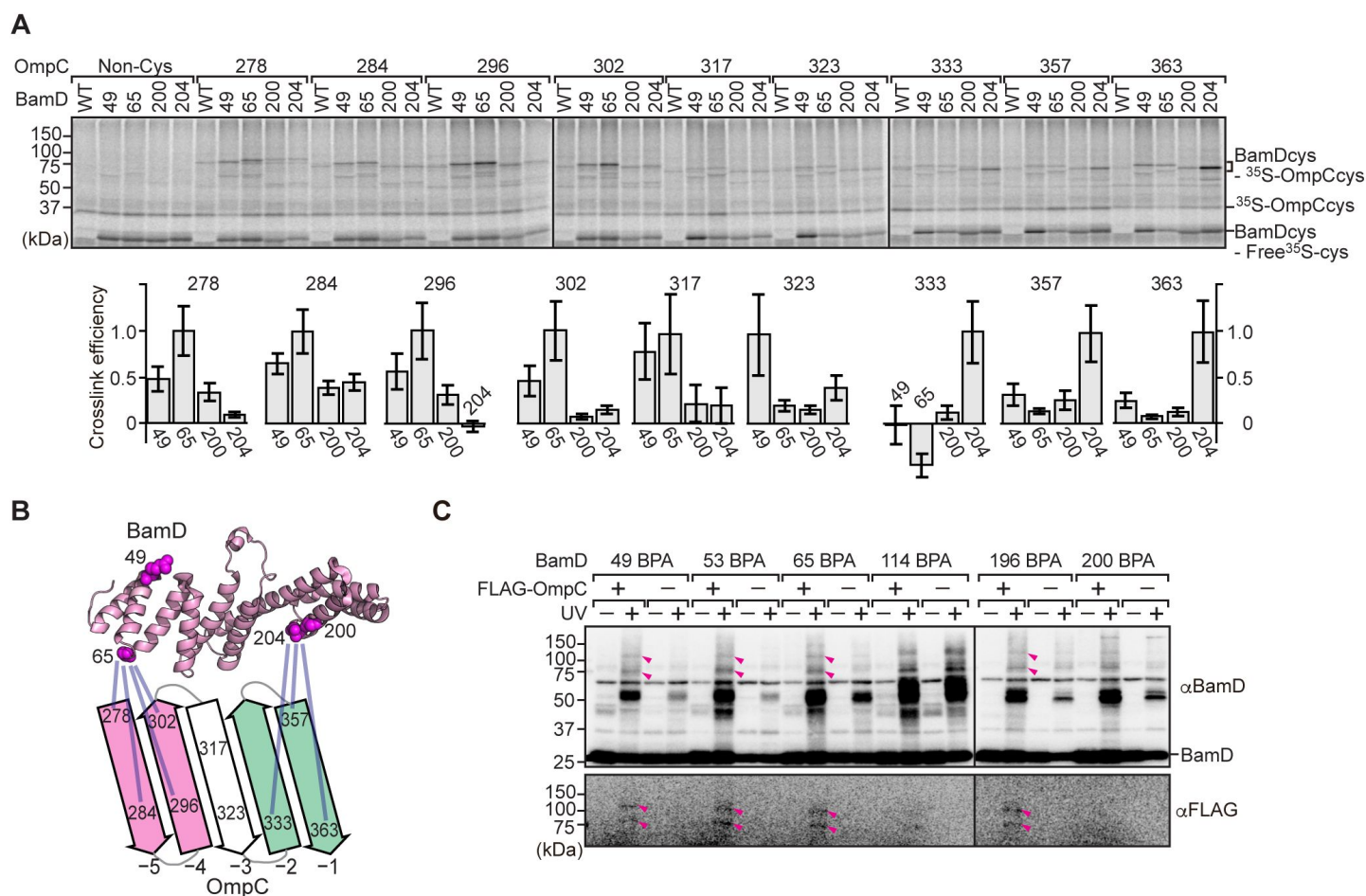


Fig. 6 |

BamD acts as the receptor for internal β -signals.

A, BamD-cys mutant proteins were purified and incubated with ^{35}S -OmpC with cysteine mutations from the -5 to the final β -strand. After the addition of the oxidizing agent, CuSO_4 , samples were purified with Ni-NTA and analyzed by SDS-PAGE. **B**, BamD cysteine mutants, highlighted in magenta, formed disulfide bonds at distinct locations of OmpC. **C**, BPA containing BamD was expressed in the presence (+) or absence (-) of FLAG-OmpC overexpression. Pink arrows indicate BamD-FLAG-OmpC cross-link products.

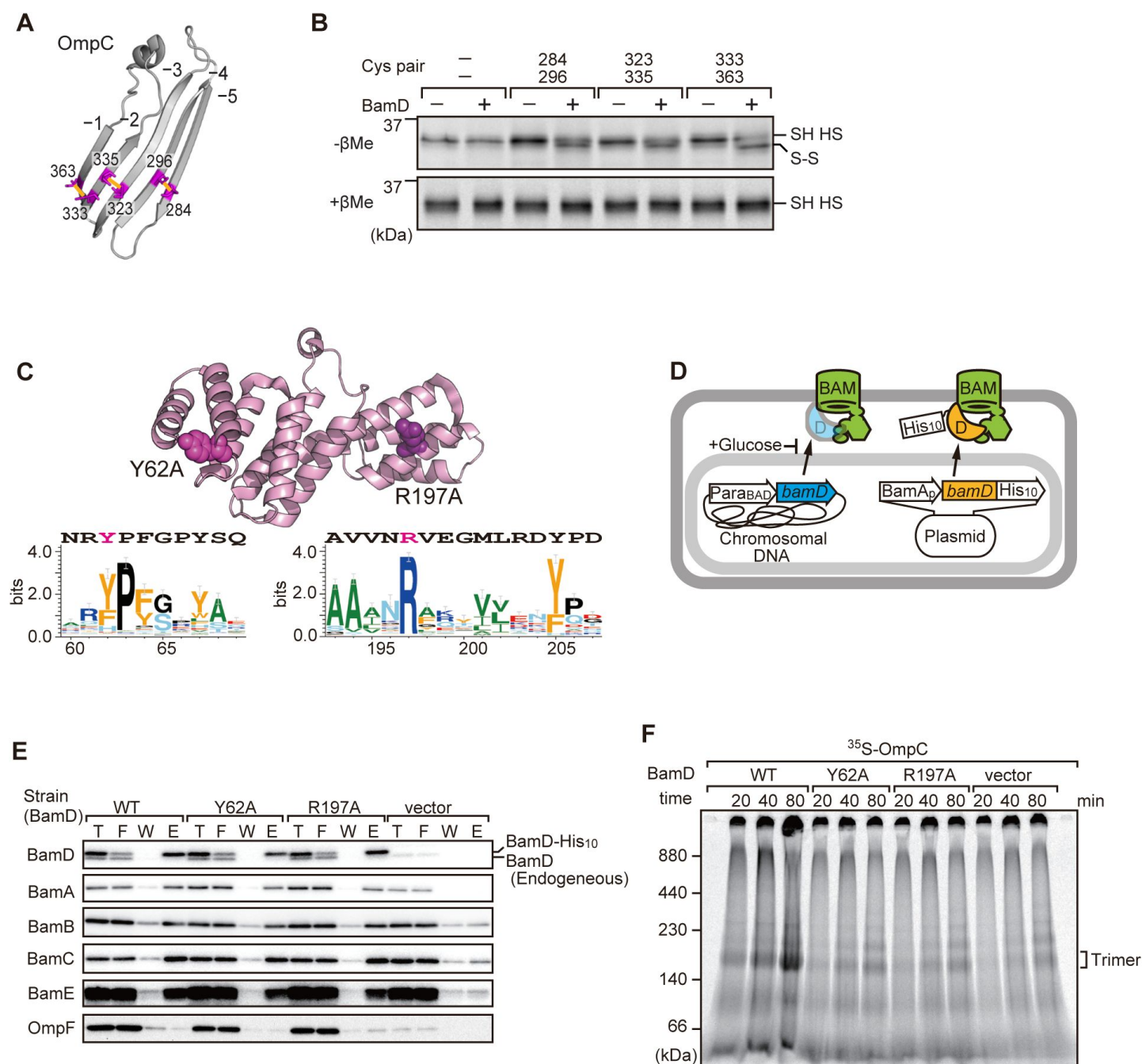


Fig. 7 |

Key residues in two structurally distinct regions of BamD promote OMP orientation and assembly.

A, Double cysteine residues were mutated into OmpC between anti-parallel β -strands to form artificial disulfide bonds **B**, ^{35}S -OmpC-cys variants were translated then incubated with (+) or without (-) BamD. The samples were analyzed by reducing or non-reducing condition. Disulfide bond specific bands (S-S) are indicated to the right of the gel. **C**, Sequence conservation analysis of BamD. Target residues, Y62 and R197, are indicated with dark pink spheres on the crystal structure of BamD, sequence logos represent conservation in the immediate region of these residues. Amino acids listed above logo sequence are from crystal structure. Sequence conservation of BamD (bottom). **D**, Schematic depiction of the BamD depletion strain of *E. coli* used to express mutant BamD proteins. **E**, Pull-down assay of the variants of BamD-His8. The EMMs as in **(C)** were solubilized with 1.5% DDM and then subjected to Ni-NTA. T-5% total input, F-unbound fraction, W-wash fraction, E-Eluted fraction. Each fraction was analyzed by SDS-PAGE and immunoblotting against indicated antibodies. **F**, Assembly of OmpC was reduced in EMMs containing BamD mutated at Y62 and R197. ^{35}S -labelled OmpC was incubated with EMM as in **(C)**, and then analyzed by BN-PAGE and radio-imaging.

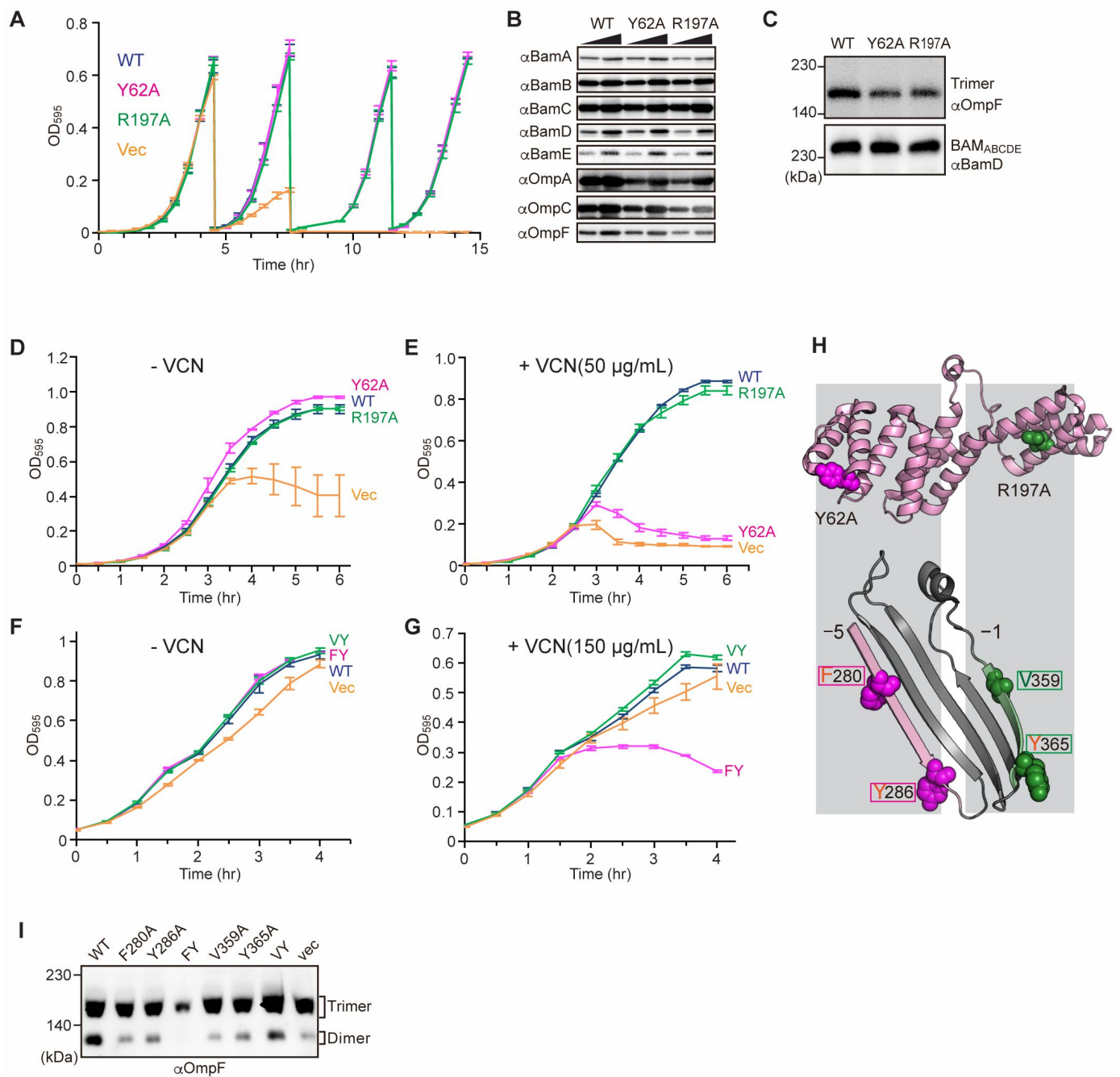


Fig. 8 |

Substrate recognition by BamD is essential to maintain membrane integrity.

A, Strains of *E. coli* with a depletion of chromosomal *bamD*, complemented by expression of the indicated forms of BamD, were grown in restrictive media (LB supplemented with 0.5% glucose) for 4 hours, then diluted into fresh media and grown a further 4 hours, then diluted into fresh media and grown a further 4 hours. **B**, Quantitation of steady-state protein levels of OmpF and the indicated subunits of the BAM complex in WT, Y62A, R197A mutants were determined after SDS-PAGE and immunoblot analysis. Wedge indicates that in each case, two samples were assessed corresponding to either 4 µg or 12 µg of total protein. **C**, Quantitation of the level of trimeric OmpF formed in the indicated EMM assays. The immunoblot analysis is shown after BN-PAGE analysis and immunoblotting with antibodies recognizing OmpF (upper panel) or BamD (lower panel). **D-G**, Growth of the indicated strains of *E. coli* in the absence (-VCN) or presence (+VCN) of the antibiotic vancomycin. **(E)** *E. coli* cells expressing mutations at two key residues in the -5 internal β-signal of OmpC were sensitive to the presence of VCN. **H**, Structure of BamD (top) and the final 5 β-strands of OmpC (bottom). Gray boxes indicate relative regions of interaction between BamD and substrates. **I**, OmpF complex assembly *in vivo* of cells expressing mutant OmpC. OmpF assembly was greatly reduced in cells expressing double mutations to the -5 internal β-signal.

Vancomycin sensitivity is a classic assay to measure membrane integrity in Gram-negative bacteria such as *E. coli* because the drug vancomycin cannot permeate the outer membrane except if outer membrane integrity is breached by any means. Then, and only then, is sensitivity to vancomycin seen (Hart and Silhavy, 2020 [↗](#)). To test whether expression of a mutant form of OmpC that cannot engage BamD *in vivo* would likewise diminish membrane integrity, we assayed for growth of *E. coli* strains expressing OmpC(FY) and other mutant forms of OmpC (Fig. 8F, 8G [↗](#)). The OmpC(FY) –5 strand double mutant showed increased sensitivity to vancomycin (Fig. 8G [↗](#)). The OmpC(FY) mutant does not accumulate in the outer membrane to the level found in wild-type *E. coli* (Fig. 3D [↗](#)). The strain co-expressing OmpC(FY) with endogenous OmpC has diminished levels of another major porin (OmpF) as judged by BN-PAGE, but not OmpC(WT) (Fig. 8I [↗](#), see also Fig. 3 [↗](#)).

Discussion

While “outer membrane integrity” is a time-honored term, what does it mean in a molecular sense? OMPs are the major components of bacterial outer membranes, accounting for more than 50% of the mass of the outer membrane (Horne et al., 2020 [↗](#); Jarosławski et al., 2009 [↗](#)). The quantitative analysis of single-cell data suggests approximately 500 molecules of major porins, such as OmpC or OmpF, are assembled into the outer membrane per minute per cell (Benn et al., 2021 [↗](#); Lithgow et al., 2023 [↗](#)). This considerable task requires factors that promote the efficient and prompt assembly of OMPs in order to handle this extraordinary rate of substrate protein flux. The efficiency of assembly is necessary to fight against external cellular insult, hence the use of BamD as an OMP assembly accelerator.

In an unbiased experimental approach to assess for inhibitory peptides, specific segments of the major porin substrate OmpC were shown to interact with the BAM complex as peptidomimetic inhibitors. Results for this experimental approach go well beyond expected outcomes. Discovery of conserved, internal signals in OMPs explains two previous reports where the –5 strand of BamA was suggested to be more similar to a β -signal motif than the C-terminal strand of BamA (Hagan et al., 2013 [↗](#); Imai et al., 2011 [↗](#)). Furthermore, a hidden Markov model analysis showed that in the sub-class of OMPs called autotransporters, which includes EspP, ~31% (473/1511) of non-redundant proteins show no such β -signal motif at the C-terminal strand, but instead contain a β -signal motif in the –5 strand (Celik et al., 2012 [↗](#); Cox et al., 2010 [↗](#)).

The inhibitory peptides discovered here contained information that led us to define an internal signal for OMP assembly. Hidden Markov model approaches like HMMER (Eddy, 1995 [↗](#)) detect conserved sequence features and tools such as MEME (Bailey et al., 2009 [↗](#)) can then define a sequence motif derived from information across very many non-redundant protein sequences. This is rich information, that has to be oversimplified when written as a primary structure description typically used for the β -signal ($\{xGxx[\Omega/\Phi]x[\Omega/\Phi]\}$) (Kutik et al., 2008 [↗](#)). Analysis of the few peptidomimetic peptide sequences in functional analyses, via both *in vitro* assays and intact *E. coli* analysis, suggests that $\Phi xxxxx[\Omega/\Phi]x[\Omega/\Phi]$ is a descriptor for the internal signal, which is complimentary in function to the C-terminal β -signal that engages the BAM complex to assist OMP assembly into the outer membrane.

The current dogma in our field states that the information contained in the β -signal is important for engagement into the lateral gate of BamA (Bitto and McKay, 2003 [↗](#); de Cock et al., 1997 [↗](#); Doyle et al., 2022 [↗](#); Doyle and Bernstein, 2019 [↗](#); Hagan et al., 2015 [↗](#); Jansen et al., 2000 [↗](#); Robert et al., 2006 [↗](#); Tomasek et al., 2020 [↗](#); Xiao et al., 2021 [↗](#)). This interaction initiates the insertion of a nascent OMP polypeptide into the outer membrane. By this model, BamD is considered to function by organizing the BamC and BamE subunits of the BAM complex against the core subunit BamA (Kim et al., 2011 [↗](#), 2007 [↗](#); Malinverni et al., 2006 [↗](#)). If that is the sole role of BamD, then BamD is essential because its importance in structural organization of the BAM

complex (Hart and Silhavy, 2020 [DOI](#)), not because of a substrate recognition function. Previous studies have shown BamD is capable of stimulating the folding and insertion of OMPs into liposomes in the absence of BamA (Hagan et al., 2013 [DOI](#), 2010 [DOI](#)). The question remains, how is BamD capable of this function? In our present study, we show that incubation of OmpC with BamD promotes the formation of antiparallel β -strands by the formation of intra-molecular disulfide cross-linking of neighboring β -strands at the C-terminus (Fig. 7B [DOI](#)). In EspP position 1214, located within the internal signal, was seen to cross-link to BamD (Ieva et al., 2011 [DOI](#)), and our inter-molecular cysteine cross-linking showed the N-terminal helices of BamD's TPR structure interacts with the internal signal and the C-terminal helices interacts with the β -signal (Fig. 6B, C [DOI](#)). Failure in this interaction was seen to stall substrate assembly within the soluble domain of the BAM complex (Fig 4C [DOI](#) and 5D [DOI](#)). This suggests that BamD acts in an important role in an early recognition step in which both the substrate's C-terminal β -signal and internal signal engage BamD in order to be arranged into a form ready for transfer to the lateral gate of BamA (Shen et al., 2023 [DOI](#); Tomasek et al., 2020 [DOI](#)) (Fig 9A [DOI](#)). The recognition and initiation of β -strand formation by BamD is necessary to support the efficient assembly of the 500 molecules of OMPs per minute required by the outer membrane, ensuring the survival against foreign insult and supports bacterial survival (Lithgow et al., 2023 [DOI](#)).

Recent advances in our understanding of BAM complex function have come from structural determination of EspP engaged in assembly by the BAM complex (Doyle et al., 2022 [DOI](#); Shen et al., 2023 [DOI](#)). The assembly intermediate of EspP engaged with the BAM complex shows the -4 strand, interacts with the R49 residue of BamD, in the N-terminal substrate binding region, even before the -5 strand had entered into this cavity (Doyle et al., 2022 [DOI](#)). Spatially, this indicates the BamD can serve to organize two distinct parts of the nascent OMP substrate within the periplasmic domain of the BAM complex, prior to, or in concert with, engagement to the lateral gate of BamA. Assessing this structure showed the N-terminal region of BamD (interacting with the POTRA1-2 region of BamA) and the C-terminal region of BamD (interacting with POTRA5 proximal to the lateral gate of BamA) (Bakelar et al., 2016 [DOI](#); Gu et al., 2016 [DOI](#); Tomasek et al., 2020 [DOI](#)) and that the N-terminal region of BamD changes conformation depending on the folding states of the last four strands of EspP. The C-terminal section of BamD and POTRA5 show no change during this stage of the folding reaction (Doyle et al., 2022 [DOI](#)), but the overall effect is that the dimensions of this cavity change, a change which is dependent on the folded state of the substrate engaged in it (Fig 9 [DOI](#) B-E). We showed that purified BamD can facilitate partial folding of an OMP and we propose that BamD captures the -1 strand and draws the -5 strand by a conformational change, thereby catalyzing the formation of hydrogen bonds to form β -hairpins in neighboring strands, to prime the substrate for insertion into the outer membrane by the action of BamA.

Materials and methods

E. coli strains and Growth conditions

E. coli strains used in this study are listed in Table S1. All strains were grown in LB (1% tryptone, 1% NaCl, and 0.5% yeast extract). *E. coli* strains used for the *in vivo* photo-crosslinking experiments were grown in XB (1% tryptone, 1% NaCl, and 0.1% yeast extract). Unless otherwise specified, ampicillin (amp; 50 mg/ml), chloramphenicol (Cm; 10 mg/ml), or kanamycin (Kan, 30 mg/ml) were added to media for plasmid selection.

To create the BamD-depletion strain, we amplified four DNA fragments encoding 540 to 40 base pairs of the 5' UTR of BamD (primers BamDSO-1 and -2), kanamycin cassette (primers BamDSO-3 and -4), AraC-pBAD (primers BamDSO-5 and -6), and the first 500 bp of BamD (primers BamSO-7 and -8), respectively (Figure S16). Table S2 describes the pair of primers and template DNA to amplify each DNA fragment. These four DNA fragments were combined by the overlap PCR

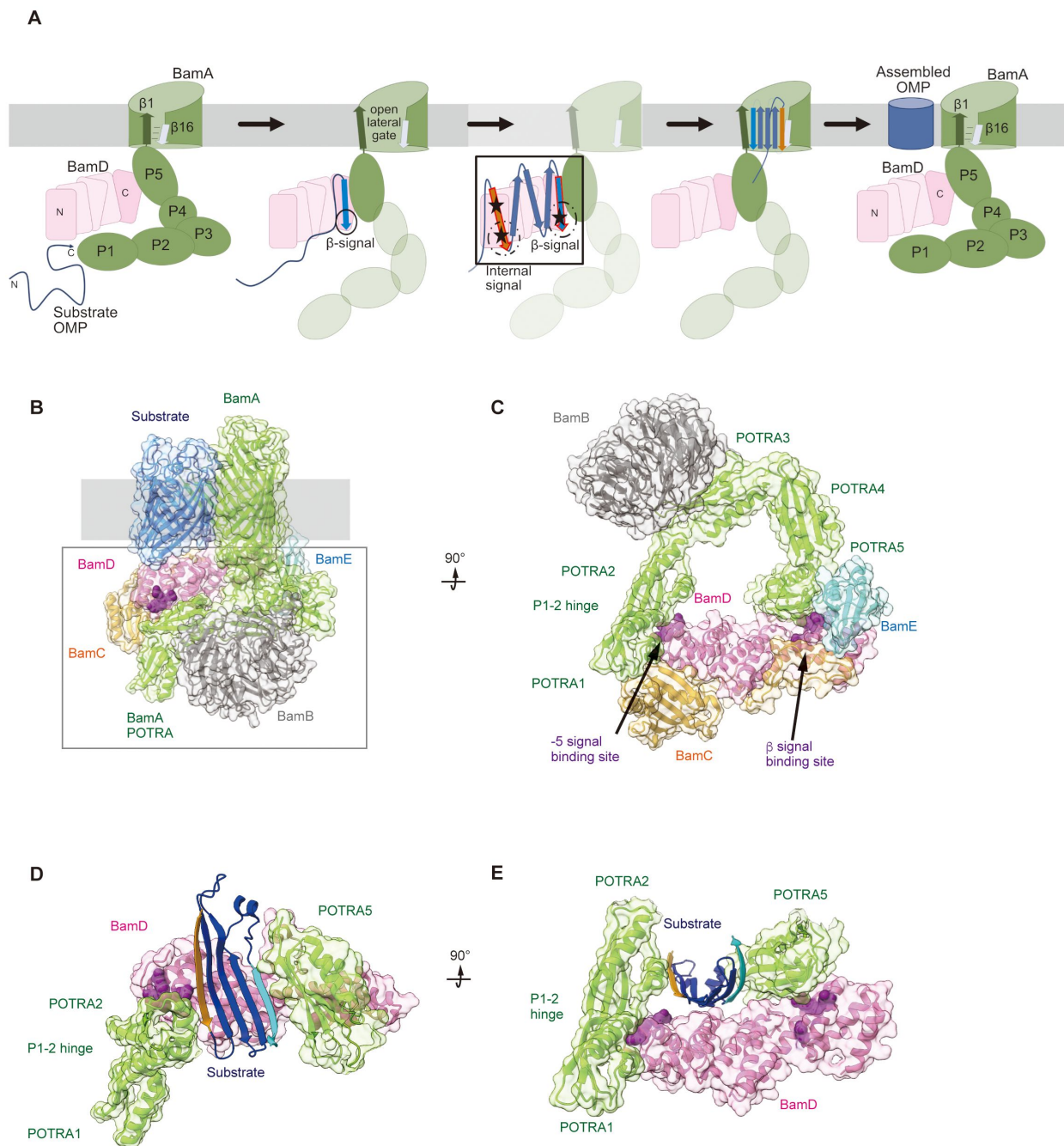


Fig. 9 |

Model for the recognition of internal and canonical β -signals by the BAM complex.

A, BAM complex assisted-OMP assembly is initiated when substrate OMPs encounter the periplasmic domain of the BAM complex. The canonical β -signal (blue arrow) of the OMP engages with the C-terminal TPR motifs of BamD (pink), after which the internal β -signal (orange arrow) is marshalled by the N-terminal TPR motif. The internal signal interacts with BamD R49 and N60 (black stars) while the canonical β -signal interacts with BamD D204 (black star). The proximity of BamD-substrate to POTRA 5 and the organization of the C-terminal β -strands of OMPs can then stimulate the entrance to the lateral gate of BamA. Finally, barrel formation and release into the membrane follows as previously published^{28,29}. **B**, Structure of substrate engaged BAM complex (PDBID: 6V05). **C**, Bottom-up view of periplasmic domain of complex as shown in **(B)**. Arrows indicate the position of the residues on BamD that function in binding sites for -5 signal and the β signal. **D**, A cavity formed by POTRA 1-2 and POTRA 5 of BamA and BamD is sufficient to accommodate the C-terminal five β -strands of OmpC. OmpC -1 strand and -5 strand are indicated by light blue strand and orange strand, respectively. The signal binding region of BamD is emphasized by deep purple (PDBID; Bam complex: 5DOQ, OmpC: 2J1N). **E**, Bottom-up view of complex as shown in **(D)**.

method and transformed into MC4100a harboring pKD46 encoding λ Red recombinase (Datsenko and Wanner, 2000). Transformants were selected in the presence of kanamycin and 0.1% (w/v) L -arabinose.

To culture BamD-depletion strains a single colony from each BamD-depletion strains harboring pBamD variants was inoculated into LB (+ amp + Kan + 0.02% (w/v) L -arabinose) and cultured for 10 h at 37°C. The saturation culture was shifted into the LB (+ amp + Kan) and incubated for another 10h at 37°C. The saturation culture was diluted into LB (+ amp + kan + 0.5% (w/v) D -Glucose) to an $\text{OD}_{600} = 0.02$, and incubated at 37°C until cell density reached $\text{OD}_{600}=0.8$. A second dilution in the same media to an $\text{OD}_{600} = 0.02$ was performed, and incubated for 2.5h at 37°C.

Plasmids

Plasmids and primers used in this study are listed in Tables S3 to S7. Cloning of specific target gene ORFs into specified vectors was performed via SLICE *in vitro* recombination (Okegawa and Motohashi, 2015).

Protein Purification

Plasmids encoding each subunit of the BAM complex and OmpC genes (including a His₆-tag) were transformed into BL21(DE3)* cells (Invitrogen). BamD and OmpC variant transformants were grown in LB (+ amp) at 37°C to an $\text{OD}_{600} = 0.5$. Expression was induced by adding 1 mM IPTG and the cells grown at 37°C for 3 h. For BamD containing BPA mutations, MC4100 cells were transformed with pHIS-BamD(X)-amber and pEVOL-BpF45. Transformants were cultured in LB with antibiotics to an $\text{OD}_{600} = 0.5$, then supplemented with 1 mM BPA, 0.05% (w/v) L -arabinose and 1 mM IPTG to induce expression. Cells were grown at 37°C for a further 3 h.

Cells were pelleted by centrifugation at 6,000 x *g* for 10 min. Pellets were resuspended in Buffer A (150 mM NaCl, 20 mM NaPO₄) and lysed via sonication. Lysates were clarified by low-speed centrifugation of 1100 x *g* for 10 min at 4°C. For purification of BamD variant proteins, supernatants were again clarified by high-speed centrifugation, 17000 x *g* for 15 min at 4°C. High-speed clarified supernatants were then applied Ni₂⁺-NTA-affinity column chromatography. Ni₂⁺-NTA resin were washed by Buffer A with 50 mM imidazole and proteins were eluted in Buffer A containing an imidazole step gradient (100 - 400 mM). Peak fractions were collected as the purified fraction. To purify OmpC, the pellets containing OmpC in inclusion bodies were resuspended in Buffer A containing 0.5% Triton X-100 and centrifuged for 11,600 x *g* for 10 min at 4°C. Inclusion bodies were denatured in Buffer A with 6 M urea and mixed at 25°C for 90 min, followed by centrifugation of 11,600 x *g* for 10 min at 4°C. The filtered supernatant was applied to Ni₂⁺-NTA-affinity column chromatography. The bound OmpC-His₆ was washed with Buffer A (+ 6 M urea, 50 mM imidazole) and eluted with Buffer A containing 6M urea and 200 mM imidazole. Purified proteins were then diluted with glycerol (final concentration of 10%), snap frozen in liquid nitrogen and stored in -80°C. BamA was purified as detailed previously (Ding et al., 2020).

Isolation of the *E.coli* microsomal membrane fraction (EMM)

E. coli crude membrane fraction were isolated as previously described (Gunasinghe et al., 2018). In brief, pelleted cells were resuspended in sonication buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl, 5 mM EDTA) and lysed via sonication on ice. After low-speed centrifugation (1,100 x *g*, 5 min, 4°C), supernatant was centrifuged again (15,000 x *g*, 10 min, 4°C). Pelleted membranes were resuspended in SEM buffer (250 mM sucrose, 10 mM MOPS-KOH pH 7.2, 1 mM EDTA) and flash frozen. Protein concentrations of EMM were calculated by OD_{280} measurements of 100 times diluted in 0.6% SDS ($\text{OD}_{280}=0.21$ was set to be 10 mg/ml).

EMM assembly assay

The EMM assembly assay has been previously described (Gunasinghe et al., 2018 [DOI](#); Thewasano et al., 2023 [DOI](#)). The method in brief is as follows; substrates were prepared via *in vitro* transcription by SP6-RNA polymerase followed by *in vitro* translation in rabbit reticulocyte lysate supplemented with ³⁵S-methionine. EMMs were resuspended in Assembly Assay buffer (10 mM MOPS-KOH pH7.2, 2.5 mM KH₂PO₄, 250 mM sucrose, 15 mM KCl, 5 mM MgCl₂, 2 mM methionine, 5 mM DTT, 1% w/v BSA, 0.09% v/v Triton X-100), and incubated with ³⁵S-labelled substrate at 30°C for indicated time points. Assembly reactions were halted by moving to ice for 5 minutes. The EMMs were then harvested via centrifugation at 15,000 x *g* for 5 min at 4°C and washed with SEM buffer. After SEM buffer wash, trimeric protein assembled EMM pellets were solubilized in 1.5% DDM containing Blue native (BN)-PAGE Lysis Buffer (25 mM imidazole-HCl pH 7.0, 50 mM NaCl, 50 mM 6-aminohexanoic acid, 1 mM EDTA, 7.5% (w/v) glycerol) on ice for 20 minutes. Solubilized proteins were clarified by centrifugation of 15,000 x *g* for 10 min at 4°C, followed by BN-PAGE analysis as described previously (Gunasinghe et al., 2018 [DOI](#); Thewasano et al., 2023 [DOI](#)).

Peptide library was synthesized by MIMOTOPES, and resuspended in DMSO at a concentration of 30 mM. The peptide inhibition screen of EspP was performed as stated above, but with the addition of 0.1 mM inhibitory peptide to the Assembly Assay buffer and incubated with EMMs for 5 minutes prior to the addition of ³⁵S-labelled substrate protein. After harvesting, EMMs were treated with 100 µg/mL Proteinase K on ice for 20 minutes. Digestion was halted by adding 2 mM PMSF. The EMMs were then collected via centrifugation at 15,000 x *g* for 5 min at 4°C and washed with SEM buffer. EspP assembled with EMM proteins were analyzed by SDS-PAGE and radio-imaging. EMM assembly of trimeric proteins was tested with the addition of 0.25 mM peptide and analyzed as above.

Sequence conservation analysis

For conservation analysis, we obtained homologous sequences of representative structure-known beta barrel outer membrane proteins from reference bacterial proteomes in UniProt (The UniProt Consortium, 2019) using phmmer and jackhammer HMMER 3.2.1 (<http://hmmer.org> [DOI](#)). The best hit sequences in organisms which also contained a BamD homolog were used. We identified 31, 94, 25, 19, 27, 44, 43, 23, 42, 22, 132, 46, 58, 68, 39, 245, 261, 115, 187, 250 and 74 sequences for OmpA, OmpW, PagP, OmpX, OmpT, EspP, Hbp, NanC, OMPLA, Tsx, FadL, OmpC, OmpF, PhoE, LamB, BtuB, Cir, FecA, FepA, FhuA and PapC, respectively (sequence identity less than 60%, expect for OmpX. In case of OmpX, sequence identity is less than 80% due to the small number of sequences). Multiple alignments were generated by MAFFT (Katoh et al., 2019 [DOI](#)) and then sequence logos of - 5th transmembrane strands were created using WebLogo 3 (Crooks, 2004 [DOI](#)).

In vivo assembly analysis of FLAG-OmpC

We introduced a FLAG epitope tag into the N-terminal region of OmpC, immediately behind the cleavage site for the SEC-signal sequence (Rapoport, 2007 [DOI](#)), to distinguish between endogenous OmpC and the OmpC variants in the OmpC-deletion strain background which has issues of diminished OM integrity. The expression of FLAG-OmpC mutants was controlled by using a plasmid-based pBADara promoter cassette, to avoid the chronic effect of constitutively expressed mutant proteins (Figure 3A [DOI](#)). We transformed pBAD-FOmpC variants into MC4100A. Transformants were cultured in LB (+ amp) until OD₆₀₀=0.5 and then protein expression induced by adding 0.1% arabinose for 3 h. *E. coli* cells were harvested and total cell lysate or EMMs prepared accordingly.

Antibody Shift Assay

The variation of BN-PAGE analysis referred to as an antibody shift assay was performed as previously described with modification (Shiota et al., 2012 [DOI](#)). In brief, solubilized EMMs were incubated with 3 µL of anti-BamC antibodies for 45 min at 4°C. Samples were clarified via

centrifugation at 13,000 x *g* for 10 minutes at 4°C, and subjected to BN-PAGE.

Urea Extraction of Intermediate

Once the Assembly Assay was performed, to obtain intermediate complexes, lysates were washed with SEM, the EMMs resuspended in SEM containing 6 M urea and incubated at 37°C for 1 hour, after which 50 mM Tris HCL pH 8.0 was added. Membrane fractions were collected via high-speed centrifugation (100,000 x *g*, 45 min, 25°C) and washed twice with SEM buffer. The membrane was solubilized for 20 minutes at 4°C in 1.5% DDM containing BN-PAGE lysis buffer and subjected to BN-PAGE analysis.

Neutron Reflectometry

Neutron reflection (NR) was carried out on the SOFIA reflectometer at the Materials and Life Science Experimental Facility (MLF) at J-PARC, Japan (Mitamura et al., 2013 [DOI](#); Yamada et al., 2011 [DOI](#)). The reflectivity at the interface between the Si substrate and water solution was measured at 3 incident angles (0.3°, 0.75° and 1.8°). The neutrons were introduced from the substrate side to illuminate the interface and the reflection intensity was normalized by the transmission intensity through the substrate to achieve the reflectivity with the *Q* range of up to 0.2 Å⁻¹.

Analysis of the NR profiles was performed using the MOTOFIT analysis software (Nelson, 2010 [DOI](#)). The software uses a conventional method, optical matrix method (Nelson, 2006 [DOI](#)) to model the reflectivity profiles. Briefly, the layer is divided into several sublayers and then a characteristic matrix is evaluated for each sublayer, from which the whole reflectivity can be calculated exactly. The same layer with multiple datasets with three different isotopic contrasts is fitted simultaneously until the satisfactory fit is obtained (smallest χ^2 value in the genetic algorithm). All models include a 15 Å oxide layer on the surface of the silicon substrates and subsequently there are three layers of Ni, gold and Ni-NTA on the top of oxide layer. Protein layers will assemble outwards from the Ni-NTA layer.

The best-fit model from MOTOFIT gives information about the thickness, density and roughness of each layer. When a layer is composed of just two components, a chemical species or its fragment *s* and water *w*, the scattering length density (SLD) is given by: where ρ_w and ρ_s are the scattering length densities of the two components, respectively and ϕ is the volume fraction of species *s* in the layer. If more than one chemical species is present, then ρ_s is determined by the volume fraction of each species in the layer.

OmpC *in vitro* folding

Urea denatured OmpC and OmpC variants were purified as stated above. Protein was resuspended in Refolding Buffer (10 mM Phosphate Buffer pH 8.0, 2 mM EDTA, 1 mM glycine) to a final concentration of 5 µM. DDM was added to a final concentration of 3.4 mM. Samples were incubated at 37°C for 0, 1, 3, 6, 8, 12, 24, and 36 hours. SDS-Loading dye was added and each sample was split into two tubes (one to be boiled, one at RT for 10min). OmpC folding was analyzed via SDS-PAGE.

Ni-NTA Substrate Pulldown

Purified BamA or BamD was resuspended in Buffer A to a final concentration of 10 µM. Specific peptides or DMSO alone was added (final concentration 0.35 mM) and incubated rocking at 4°C for 1 hour. To each tube, 0.5 µL of ³⁵S-OmpC was added and vortexed vigorously, 10% of the total volume was removed to assay as “input” for each sample. Samples were then incubated at 4°C with rocking for 30 minutes. Ni-NTA agarose beads were add to each sample, incubated at 4°C with

rocking for an additional 30 minutes then washed with Buffer A containing 20 mM imidazole. BAM proteins were eluted from the Ni-NTA resin using Buffer A with 400 mM imidazole. Proteins were TCA precipitated and subjected to analysis by SDS-PAGE and imaged via radio-imaging.

Disulfide cross-linking

BamD-OmpC cysteine crosslinking was performed using 50 μ M purified single-cysteine BamD mutant variants incubated with 8 μ L of single-cysteine OmpC variants translated by the retic in 30 μ L of Buffer A at 25°C for 2 min. An oxidizing reaction mixture of 1 μ M CuSO₄, 4 mM methionine, and 4 mM cysteine was added and incubated at 25°C for 5 min. Proteins were precipitated by TCA precipitation, and solubilized in 1% SDS buffer. Solubilized proteins were diluted by 0.5% TritonX-100 buffer and purified with Ni-NTA via the tagged BamD-H6. BamD and disulfide crosslink products were eluted with 400 mM imidazole containing 0.5% TritonX-100 buffer. Eluted proteins were concentrated by TCA precipitation and solubilized with SDS-PAGE loading dye with or without β -Me.

The OmpC intrachain disulfide bond assay was performed using 1 μ L ³⁵S-OmpC double cysteine variants translated in the presence of 1 mM DTT. OmpC variants were diluted into 100 μ L Buffer A then incubated with 50 μ M BamD at 32°C for 30 minutes. 10 μ M CuSO₄ was added to samples and incubated for an additional 2 minutes, followed by the addition of SDS-PAGE loading dye with or without 5mM DTT and 1% β -Me. Proteins were resolved via SDS-PAGE and imaged as stated above.

In vivo and *in vitro* photo-crosslinking

For *in vivo* photo-crosslinking, MC4100 strain was transformed with pEVOL-pBpF, pTnT-BamAp-BamD-His8, and pET-FLAG-OmpC, with respective antibiotics. A single colony was used to inoculate a culture in photo-crosslinking media (1% NaCl, 1% Tryptone, 0.1% Extract yeast dried, 1mM BPA) for 3 hours at 37°C in the dark. Afterward, arabinose (0.1% final concentration) was added and further incubated for 2.5 h. Cultures were divided with one half UV-irradiated for 7 min at RT and the other half not. Cells were pelleted and solubilized with 1% SDS buffer (50mM Tris-HCl pH 8.0, 150mM NaCl, 1% SDS). His-tagged BamD containing BPA and its cross-linked products were purified by Ni-NTA agarose. Cross-linked products were identified via immunoblotting with anti-BamD and anti-FLAG antibodies.

In vitro photo-crosslinking was performed with 32 μ g of purified BamD-His₈ variants containing BPA diluted in Buffer A. 40 μ L of ³⁵S-OmpC was added to each tube and incubated at 25°C for 10 minutes in the dark. Each mixture was transferred to the lid of a microtube and irradiated with UV using B-100AP (UVP) for 7 min on ice. 10% (w/v) SDS solution was added to a final concentration of 1%, then incubated at 95°C for 10 min to denature proteins. Protein mixtures were diluted with 0.5% TritonX-100 buffer (50 mM Tris-HCl pH 7.0, 150 mM NaCl, 0.5% Triton-X100), and purified on Ni-NTA agarose beads. Elute fractions were concentrated by TCA precipitation and analyzed by SDS-PAGE and radio-imaging.

Statistical analysis

For all densitometry measurements we utilized “imagequant” quant (ImageQuant TL, cytiva) and statistical analysis and graphing was performed with Microsoft Excel software.

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Competing interests

Authors declare that they have no competing interests.

Data and materials availability

All data are available in the main text or the supplementary materials.

Supplementary materials

This PDF file includes

Legends for Supplementary figures. Figs. S1 to S16

Tables S1 to S7

References for Supplementary materials

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Reviewer #1 (Public Review):

Summary:

The biogenesis of outer membrane proteins (OMPs) into the outer membranes of Gram-negative bacteria is still not fully understood, particularly substrate recognition and insertion by beta-assembly machinery (BAM). In the studies, the authors present their studies that in

addition to recognition by the last strand of an OMP, sometimes referred to as the beta-signal, an additional signal upstream of the last strand is also important for OMP biogenesis.

Strengths:

1. Overall the manuscript is well organized and written, and addresses an important question in the field. The idea that BAM recognizes multiple signals on OMPs has been presented previously, however, it was not fully tested.
2. The authors here re-address this idea and propose that it is a more general mechanism used by BAM for OMP biogenesis.
3. The notion that additional signals assist in biogenesis is an important concept that indeed needs fully tested in OMP biogenesis.
4. A significant study was performed with extensive experiments reported in an attempt to address this important question in the field.
5. The identification of important crosslinks and regions of substrates and Bam proteins that interact during biogenesis is an important contribution that gives clues to the path substrates take en route to the membrane.

Weaknesses:

Major critiques (in no particular order):

1. The title indicates 'simultaneous recognition', however no experiments were presented that test the order of interactions during OMP biogenesis.
2. Aspects of the study focus on the peptides that appear to inhibit OmpC assembly, but should also include an analysis of the peptides that do not to determine this the motif(s) present still or not.
3. The b-signal is known to form a b-strand, therefore it is unclear why the authors did not choose to chop OmpC up according to its strands, rather than by a fixed peptide size. What was the rationale for how the peptide lengths were chosen since many of them partially overlap known strands, and only partially (2 residues) overlap each other? It may not be too surprising that most of the inhibitory peptides consist of full strands (#4, 10, 21, 23).
4. It would be good to have an idea of the propensity of the chosen peptides to form b-stands and participate in b-augmentation. We know from previous studies with darobactin and other peptides that they can inhibit OMP assembly by competing with substrates.
5. The recognition motifs that the authors present span up to 9 residues which would suggest a relatively large binding surface, however, the structures of these regions are not large enough to accommodate these large peptides.
6. The authors highlight that the sequence motifs are common among the inhibiting peptides, but do not test if this is a necessary motif to mediate the interactions. It would have been good to see if a library of non-OMP related peptides that match this motif could also inhibit or not.
7. In the studies that disrupt the motifs by mutagenesis, an effect was observed and attributed to disruption of the interaction of the 'internal signal'. However, the literature is filled with point mutations in OMPs that disrupt biogenesis, particular those within the membrane region. F280, Y286, V359, and Y365 are all residues that are in the membrane region that point into the membrane. Therefore, more work is needed to confirm that these mutations are in parts of a recognition motif rather than on the residues that are disrupting stability/assembly into the membrane.

8. The title of Figure 3 indicates that disrupting the internal signal motif disrupts OMP assembly, however, the point mutations did not seem to have any effect. Only when both 280 and 286 were mutated was an effect observed. And even then, the trimer appeared to form just fine, albeit at reduced levels, indicating assembly is just fine, rather the rate of biogenesis is being affected.

9. In Figure 4, the authors attempt to quantify their blots. However, this seems to be a difficult task given the lack of quality of the blots and the spread of the intended signals, particularly of the 'int' bands. However, the more disturbing trend is the obvious reduction in signal from the post-urea treatment, even for the WT samples. The authors are using urea washes to indicate removal of only stalled substrates. However a reduction of signal is also observed for the WT. The authors should quantify this blot as well, but it is clear visually that both WT and the mutant have obvious reductions in the observable signals. Further, this data seems to conflict with Fig 3D where no noticeable difference in OmpC assembly was observed between WT and Y286A, why is this the case?

10. The pull down assays with BamA and BamD should include a no protein control at the least to confirm there is no non-specific binding to the resin. Also, no detergent was mentioned as part of the pull downs that contained BamA or OmpC, nor was it detailed if OmpC was urea solubilized.

11. The neutron reflectometry experiments are not convincing primarily due to the lack controls to confirm a consistent uniform bilayer is being formed and even if so, uniform orientations of the BamA molecules across the surface. Further, no controls were performed with BamD alone, or with OmpC alone, and it is hard to understand how the method can discriminate between an actual BamA/BamD complex versus BamA and BamD individually being located at the membrane surface without forming an actual complex. Previous studies have reported difficulty in preparing a complex with BamA and BamD from purified components. Additionally, little signal differences were observed for the addition of OmpC. However, an elongated unfolded polypeptide that is nearly 400 residues long would be expected to produce a large distinct signal given that only the C-terminal portion is supposedly anchored to BAM, while the rest would be extended out above the surface. The depiction in Figure 5D is quite misleading when viewing the full structures on the same scales with one another.

12. In the crosslinking studies, the authors show 17 crosslinking sites (43% of all tested) on BamD crosslinked with OmpC. Given that the authors are presenting specific interactions between the two proteins, this is worrisome as the crosslinks were found across the entire surface of BamD. How do the authors explain this? Are all these specific or non-specific?

13. The study in Figure 6 focuses on defined regions within the OmpC sequence, but a more broad range is necessary to demonstrate specificity to these regions vs binding to other regions of the sequence as well. If the authors wish to demonstrate a specific interaction to this motif, they need to show no binding to other regions.

14. The levels of the crosslinks are barely detectable via western blot analysis. If the interactions between the two surfaces are required, why are the levels for most of the blots so low?

15. Figure 7 indicates that two regions of BamD promote OMP orientation and assembly, however, none of the experiments appears to measure OMP orientation? Also, one common observation from panel F was that not only was the trimer reduced, but also the monomer. But even then, still a percentage of the trimer is formed, not a complete loss.

16. The experiment in Fig 7B would be more conclusive if it was repeated with both the Y62A and R197A mutants and a double mutant. These controls would also help resolve any effect

from crowding that may also promote the crosslinks. Further, the mutation of R197 is an odd choice given that this residue has been studied previously and was found to mediate a salt bridge with BamA. How was this resolved by the authors in choosing this site since it was not one of the original crosslinking sites?

17. As demonstrated by the authors in Fig 8, the mutations in BamD lead to reduction in OMP levels for more than just OmpC and issues with the membrane are clearly observable with Y62A, although not with R197A in the presence of VCN. The authors should also test with rifampicin which is smaller and would monitor even more subtle issues with the membrane. Oddly, no growth was observed for the Vec control in the lower concentration of VCN, but was near WT levels for 3 times VCN, how is this explained?

18. While Fig 8I indeed shows diminished levels for FY as stated, little difference was observed for the trimer for the other mutants compared to WT, although differences were observed for the dimer. Interestingly, the VY mutant has nearly WT levels of dimer. What do the authors postulate is going on here with the dimer to trimer transition? How do the levels of monomer compare, which is not shown?

19. In the discussion, the authors indicate they have '...defined an internal signal for OMP assembly', however, their study is limited and only investigates a specific region of OmpC. More is needed to definitively say this for even OmpC, and even more so to indicate this is a general feature for all OMPs.

20. In the proposed model in Fig 9, it is hard to conceive how 5 strands will form along BamD given the limited surface area and tight space beneath BAM. More concerning is that the two proposal interaction sites on BamD, Y62 and R197, are on opposite sides of the BamD structure, not along the same interface, which makes this model even more unlikely. As evidence against this model, in Figure 9E, the two indicates sites of BamD are not even in close proximity of the modeled substrate strands.

Reviewer #2 (Public Review):

Previously, using bioinformatics study, authors have identified potential sequence motifs that are common to a large subset of beta-barrel outer membrane proteins in gram negative bacteria. Interestingly, in that study, some of those motifs are located in the internal strands of barrels (not near the termini), in addition to the well-known "beta-signal" motif in the C-terminal region.

Here, the authors carried out rigorous biochemical, biophysical, and genetic studies to prove that the newly identified internal motifs are critical to the assembly of outer membrane proteins and the interaction with the BAM complex. The author's approaches are rigorous and comprehensive, whose results reasonably well support the conclusions. While overall enthusiastic, I have some scientific concerns with the rationale of the neutron refractory study, and the distinction between "the intrinsic impairment of the barrel" vs "the impairment of interaction with BAM" that the internal signal may play a role in. I hope that the authors will be able to address this.

Strengths:

1. It is impressive that the authors took multi-faceted approaches using the assays on reconstituted, cell-based, and population-level (growth) systems.
2. Assessing the role of the internal motifs in the assembly of model OMPs in the absence and presence of BAM machinery was a nice approach for a precise definition of the role.

Weaknesses:

1. The result section employing the neutron refractory (NR) needs to be clarified and strengthened in the main text (from line 226). In the current form, the NR result seems not so convincing.

What is the rationale of the approach using NR?

What is the molecular event (readout) that the method detects?

What are "R"-y axis and "Q"-x axis and their physical meanings (Fig. 5b)?

How are the "layers" defined from the plot (Fig. 5b)?

What are the meanings of "thickness" and "roughness" (Fig. 5c)?

What are the meanings of the increases in thickness and roughness?

What does "SLD" stand for?

2. In the result section, "The internal signal is necessary for insertion step of assembly into OM"

This section presents an important result that the internal beta-signal is critical to the intrinsic propensity of barrel formation, distinct from the recognition by BAM complex. However, this point is not elaborated in this section. For example, what is the role of these critical residues in the barrel structure formation? That is, are they involved in any special tertiary contacts in the structure or in membrane anchoring of the nascent polypeptide chains?